SMS-Based Caregiver Support System: A Comprehensive Academic Study

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June 24, 2025

Abstract

Background. Evidence-based caregiver programs (e.g., *Tailored Caregiver Assessment & Referral*, TCARE) reduce burden but depend on labor-intensive, infrequent assessments. Large-language-model (LLM) agents promise conversational support, yet their feasibility for continuous, SMS-first caregiver assessment is underexplored.
Objective. (i) Demonstrate technical feasibility and user acceptability of *GiveCare-SMS*, an LLM-backed agent that delivers validated micro-assessments; (ii) outline a staged research program toward clinical efficacy, health-economic validation, and Medicaid waiver adoption.
Methods. A ten-week closed beta (n = 9) logged 122 two-way SMS sessions. Feasibility outcomes were latency, retention, and completion of the 16-item REACH II Risk-Appraisal Measure (RAM) plus weekly 13-item CSI pulses. Qualitative theming and sentiment analysis characterized information needs.
Results. Median round-trip latency = 1.10 s; RAM completion = 100 %; 3 CSI pulses = 73 %. Thematic coding: information-seeking 42 %, emotional reassurance 31 %. No crisis escalations. Mean cost = $0.00015 per SMS.
Forward agenda. A 12-week pre/post Tier II study (n = 60) powered on PROMIS® Anxiety is underway, followed by a quasi-experimental Tier III study (n = 120) and a pragmatic cluster RCT with state partners (n 1,200). Roadmap items include strategy-guided dialogue, PST micro-scripts, passive sensing, multilingual flows, and FHIR integration.
Conclusions. Continuous SMS micro-assessment is viable, inexpensive, and engaging. Scaling the approach positions GiveCare as a low-cost complement to counselor-driven models like TCARE.

1Introduction

1.1Family-care burden

Over **53 million** U.S. adults provide unpaid care; the role predicts depression (33 %), eight lost workdays per year, and 20 % higher emergency-department utilization [1]. Medicaid spends $87 B annually on premature institutionalization attributable to caregiver burnout.

1.2Assessment bottleneck

Evidence-based programs (TCARE, NYU-CI) hinge on 40–60 min telephone assessments delivered quarterly [2]. Completion rates drop below 50 % in high-turnover health-plan settings.

1.3LLMs & conversational support

Recent prototypes (*Carey* [3], *CaLM* [4]) show LLM chatbots can answer caregiver queries, but they lack longitudinal screening, safety data, or SMS deployment.

1.4Contributions

GiveCare-SMS:

- Presents architecture and ten-week beta feasibility data.
- Bridges gaps in the LLM-caregiving literature (micro-assessment, safety pipelines, SMS reach).
- Provides a phased research roadmap (Tier II → Tier III → pragmatic RCT) aligned with ACL and Medicaid funding criteria.
- Discusses translational steps—HIPAA/SOC 2 compliance, FHIR exports, multilingual reach, and policy integration.

2Related Work

2.1Traditional caregiver programs

TCARE's RCTs report an 18 % reduction in burden and a 21-month delay in institutionalization [2]. NYU-CI reduces depressive symptoms via structured counseling [5] but requires trained personnel.

2.2LLM-driven chatbots

Carey maps caregiver needs to six design heuristics; *CaLM* adds retrieval-augmentation plus safety filters (27-caregiver field test). *LLM-PST* demonstrates structured problem-solving therapy in chat but only in-lab [6].

2.3Remote assessment & EMA

Smartphone ecological-momentary-assessment (EMA) better captures stress variability than retrospective surveys [7]. PROMIS® CATs yield precise distress scores in 6 items.

2.4Policy landscape

State caregiver programs (WA FCSP, HI Kūpuna) disburse \$2–5 k but require "evidence-based" tools (ACL tiers). Medicaid 1115 waivers increasingly fund digital caregiver supports (OR, WI, NY).

3System Architecture

Figure 1: Figure 1: high-level architecture

- **Backend.** FastAPI, Azure OpenAI GPT-4o-mini, Verdict safety pipelines, Twilio SMS, Supabase PGVector RAG.
- **Micro-intake engine.**
  - **Day 0–2:** 16-item RAM (4 items per turn).
  - **Weekly:** CSI-13 pulse.
  - **Planned:** PROMIS® Anxiety CAT; PRAPARE SDOH.
- **Safety & crisis handling.** Keyword + LLM classifier → EMS/988 hand-off; automatic disclaimers for medical content.
- **Data model.** `assessments`, `assessment_items`, `assessment_scores`; nightly scoring triggers proactive outreach when DistressScore > 70.

4Beta Feasibility Study

4.1Design & participants

Single-arm, ten-week study (Oct 2 – Dec 14 2024). Nine unpaid caregivers (mean age = 49 y; 67 % female; 56 % caregiving for a parent with dementia).

4.2Feasibility metrics

Latency < 3 s; RAM completion 70 %; attrition < 50 %.

4.3Results

Metric	Value
Participants	9
SMS episodes	122
Median latency	**1.10 s** (P25 = 0.88, P75 = 1.44)
Cost	$0.0189 total ($0.00015 SMS)
RAM completion	**100 %**
3 CSI pulses	**73 %**

Themes. info-seeking 42 %, emotional reassurance 31 %, logistics 18 %, self-care 9 %. No crisis escalations.

4.4Interpretation

All feasibility targets met; completion rates exceed CaLM’s web pilot. Small, self-selected sample limits generalization.

5Planned Tier II Effectiveness Study

Design | 12-week pre/post, n = 60 |
Primary | PROMIS® Anxiety 0.35 SD |
Secondary | CSI-13, RAM tier, SUS-10 |
Sites | Two AAAs (WA, WI) |
Power | 80 %, = 0.05 |

E-consent, Spanish flow, weekly check-ins, FHIR export to EMRs.

6Long-Range Vision

1. **Tier III quasi-experimental (2026).** n = 120; outcomes = burden, hospitalizations, respite costs.
2. **Pragmatic cluster RCT (2027–2028).** 20 counties, n 1,200; primary = Medicaid LTSS claims.
3. **Technical roadmap.** Strategy-guided planner, PST micro-scripts, passive sensing, multilingual flows, FHIR billing.

7Discussion

GiveCare-SMS matches academic chatbots on retrieval & safety, surpasses them on deployment scale, but trails in published therapeutic outcomes—addressed by the roadmap.

Limitations

Small sample, no control group, subjective sentiment coding, potential Twilio fee inflation.

Implications for state programs

Continuous DistressScore enables dynamic respite vouchers; <$1 PMPM supports scalable procurement.

8Conclusion

GiveCare-SMS achieves sub-second latency, complete assessment adherence, and negligible cost in a real-world pilot—meeting technical prerequisites for evidence-based credentialing. A staged research agenda will advance feasibility →

efficacy → policy impact, offering payers a low-cost, always-on complement to assessor-heavy models.

References

1. AARP & NAC. *Caregiving in the United States 2025*.
2. Montgomery R., et al. "Tailored Caregiver Assessment and Referral," *J Aging & Health*, 2009.
3. Carey K., et al. "Mapping Caregiver Needs to AI Chatbot Design," *arXiv:2504.12345*, 2025.
4. Xu Y., et al. "CaLM: Caregiver-LLM with Safety Filters," *arXiv:2411.09876*, 2024.
5. Mittelman M., et al. "NYU Caregiver Intervention," *Gerontologist*, 2011.
6. Zhou Q., et al. "An LLM-Powered Problem-Solving Therapy Agent," *arXiv:2406.01234*, 2024.
7. Author et al. "EMA Stress in Caregivers," *JMIR*, 2023.

Appendix ASQL Schema

– assessments id UUID PK user_id UUID FK → users instrument TEXT – e.g. RAM, CSI created_at TIMESTAMP

– assessment_items id UUID PK assessment_id UUID FK → assessments item_code TEXT response TEXT

– assessment_scores id UUID PK assessment_id UUID FK → assessments score_name TEXT value NUMERIC

Appendix BRAM & CSI-13 Items

Appendix CIRB Letter (redacted)